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BATTERY PACK REPLACEMENT SYSTEM FOR VEHICLES

Abstract

A battery pack replacement system for vehicles is described, including a plurality of first passages, at least one second passage, and at least one processing equipment. The vehicles move along their respective first passages, and the second passage is arranged to intersect with these first passages in a projection direction. The intersection positions of the second passage with these first passages each have a processing area. The processing equipment moves along the corresponding second passage between these processing areas, performing a battery pack replacement operation on vehicles located in each processing area. The battery pack replacement operation involves the processing equipment unloading a used battery pack from the vehicle and loading a charged battery pack onto the vehicle.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a processing system, in particular to a battery pack replacement system applied for use in vehicles.

BACKGROUND TECHNOLOGY

[0002] The operation of electric vehicles is driven by the electrical energy stored in the battery pack. Therefore, maintaining the energy level of the battery pack becomes one of the primary tasks. In general, there are at least three methods available to maintain the energy level of the battery pack in the electric vehicle (EV): the first involves alternating current (AC) charging of the battery pack installed in the EV (commonly known as slow charging), the second involves direct current (DC) charging of the battery pack installed in the EV (commonly known as fast charging), and the third involves the direct replacement of the battery pack mounted on the EV.

SUMMARY OF THE INVENTION

Technical Problem

[0003] The primary drawback of replenishing the energy of the battery pack through charging is the time consumption. For instance, to charge 80% of the energy using slow charging may take approximately 3 to 5 hours, and even with fast charging, it still requires at least 20 to 30 minutes. Therefore, the use of charging methods makes it challenging to replenish the energy of the battery pack instantly and rapidly, resulting in inconvenience in practical applications.

[0004] Compared to replenishing energy through charging, directly replacing the battery pack allows the EV to immediately obtain the required energy. However, due to the proportional relationship between the weight and energy density of the battery pack, larger battery capacities result in heavier weights, posing challenges for manual execution of the battery pack replacement operation. In current technology, the battery pack of a typical EV constitutes approximately one-fourth of the total vehicle weight. Taking a 2000 kg EV as an example, the weight of its battery pack is around 500 kg. Therefore, automated replacement equipment is necessary for replacing the battery pack. Additionally, the batteries used for replacement cannot be shared among different EV models, further complicating the implementation of battery pack replacement.

[0005] Although replacing the battery pack allows the EV to replenish energy more quickly, the cost of establishing automated replacement equipment is at least tens of times higher than that of fast charging devices, thereby impacting the scale of implementation. Insufficient replacement equipment cannot meet the demand of most vehicles during peak hours, leading to owners having to queue for replacement. On the other hand, if an excessive number of replacement units are installed, it will result in a substantial capital investment, especially considering the low utilization rate during off-peak periods for battery pack replacements. This scenario is particularly relevant in certain locations such as rest areas along long-distance or highway routes, where there is a significant demand for battery pack replacements during holidays or weekends (i.e., peak demand), while demand is lower on regular weekdays (i.e., off-peak demand).

[0006] Furthermore, the battery pack replacement operation falls within the realm of the service industry, and its product combination cannot be pre-planned. It is necessary to choose the corresponding battery pack and solution combination based on the requirement of the EV after it arrives before proceeding with the replacement operation. Any delays in operation due to this process will severely impact the overall operational efficiency, posing disadvantages for both service providers and consumers. Moreover, current battery pack replacement operations can only be performed on a single brand or a single specification of EV. This inconvenience restricts

consumer purchasing and usage intentions regarding battery replacements.

[0007] Therefore, while replenishing the energy of the EV through battery pack replacement is a preferable option, achieving a balance between the investment cost, utilization rate, operational efficiency, and overall efficiency of the battery pack replacement system is challenging, especially considering the peak and off-peak characteristics mentioned above. In response to these challenges, the inventor has provided a battery pack replacement system for vehicles to address the aforementioned issues.

Solution to the Problem

Technical Solution

[0008] The objective of the present invention is to provide a battery pack replacement system for vehicles that can reduce the unit cost of replacing the battery pack for one vehicle, enhance system interoperability, and support flexible system expansion.

[0009] To achieve the above objectives, the present invention provides a battery pack replacement system for vehicles, including a plurality of first passages, at least one second passage, and at least one processing equipment. The first passages are each designed to facilitate the movement of at least one vehicle. The second passage is arranged to intersect with these first passages in a projection direction, and the overlapping positions of the second passage with these first passages each have a processing area. The processing equipment moves along the corresponding second passage between these processing areas, and performs a battery pack replacement operation on at least one of the vehicles in one of these processing areas. The battery pack replacement operation involves the processing equipment unloading a used battery pack from the vehicle and loading a charged battery pack onto the vehicle.

[0010] In one embodiment, the battery pack replacement system also includes a battery pack storage area, situated adjacent to the second passage for storing a plurality of battery packs. The battery packs include a used battery pack, a charged battery pack, and combinations thereof. Additionally, the processing equipment moves between the battery pack storage area and the various processing areas to transport both the used battery packs and the charged battery packs.

[0011] In one embodiment, the battery pack replacement system also includes a battery pack storage area and a battery pack conveyor device. The battery pack storage area is situated adjacent to the second passage and stores a plurality of battery packs, including the used battery pack, the charged battery pack, and combinations thereof. The battery pack conveyor device moves along the second passage between the battery pack storage area and the processing equipment. The battery pack conveyor device is designed to access the battery packs at either the battery pack storage area or the corresponding position relative to the processing equipment.

[0012] In one embodiment, the battery pack replacement system also includes a third passage, a battery pack storage area, and a battery pack conveyor device. The third passage is located on one side of the second passage. The battery pack storage area is adjacent to the third passage and stores a plurality of battery packs, including the used battery pack, the charged battery pack, and combinations thereof. The battery pack conveyor device moves along the third passage between the battery pack storage area and the corresponding position relative to the processing equipment. The battery pack conveyor device is designed to access the battery packs at either the battery pack storage area or the corresponding position relative to the processing equipment.

[0013] In one embodiment, the battery pack conveyor device accesses the battery packs while both itself and the processing equipment are in motion.

[0014] In one embodiment, the battery pack replacement system also includes a battery pack conveyor device, which moves along the second passage between the battery pack storage area and each processing area to transport the used battery pack, the charged battery pack, and their combinations.

[0015] In one embodiment, the second passage has two intersection positions with each of the first passages. In another embodiment, the second passage is circularly arranged with the battery pack

storage area as the center.

[0016] In one embodiment, the third passage is located between two adjacent second passages.

[0017] In one embodiment, the composition of the battery pack replacement operation performed in each processing area is varied. It includes different vehicle models, varying battery pack capacities, diverse quantities of battery packs, and a variety of combinations thereof.

[0018] In one embodiment, the number of processing equipment moving along the second passage is more than one.

[0019] In one embodiment, when the number of processing equipment is more than one, then at least two of the processing equipment have different applicable vehicle specifications or applicable battery pack specifications.

[0020] In one embodiment, the battery pack replacement operation also involves unloading and/or loading a plurality of battery packs for the same vehicle.

[0021] In addition, to achieve the above objectives, the present invention also provides a battery pack replacement system for vehicles, including a first passage, a second passage, and at least one processing equipment. The first passage is capable of accommodating a plurality of vehicles simultaneously or allowing them to move within it. The second passage has a plurality of overlapping positions with the first passage in a projection direction, and each overlapping position has a processing area. The processing equipment moves along the second passage between the processing areas and performs a battery pack replacement operation on different vehicles in different processing areas. The battery pack replacement operation includes the processing equipment unloading a used battery pack from the vehicle and loading a charged battery pack onto the vehicle.

[0022] In one embodiment, the battery pack replacement system also includes a third passage, a battery pack storage area, and a battery pack conveyor device. The third passage is located on one side of the second passage. The battery pack storage area is adjacent to the third passage and stores a plurality of battery packs. The battery pack conveyor device moves along the third passage between the battery pack storage area and the corresponding position relative to the processing equipment. The battery pack conveyor device accesses the battery packs at the battery pack storage area and/or the corresponding position relative to the processing equipment.

[0023] As mentioned above, the battery pack replacement system for vehicles according to the present invention has at least the following three features:

[0024] Feature 1: In the three-dimensional structure, the second passage spans across the first passage, presenting an overlapping arrangement in the projection direction. The battery pack replacement operation is executed in this three-dimensional structure. The processing equipment utilizes the movement and positioning idle time of vehicles on the first passages to perform the battery pack replacement operation on other vehicles that have already been positioned on different first passages. This enhances the operational efficiency of the high-value processing equipment, thereby reducing the unit cost of replacing the vehicle's battery pack.

[0025] Feature 2: When the number of processing equipment is more than one, the applicable vehicle specifications or applicable battery pack specifications for each processing equipment can be different. The processing equipment and the vehicles prepared for the battery pack replacement operation move in different passages and different planes, allowing flexible allocation of positions for performing the battery pack replacement operation. This enables vehicles with different specifications to execute the battery pack replacement operation within the same battery pack replacement system, enhancing system interoperability.

[0026] Feature 3: A single processing equipment can perform the battery pack replacement operation on different vehicles on different first passages. Additionally, when the battery pack replacement system needs expansion, it is possible to independently add first passages or independently add processing equipment, allowing flexible system expansion.

Advantages of the Invention

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. **1** illustrates a schematic diagram of the battery pack replacement system for vehicles in the first embodiment of the present invention.

[0028] FIG. **2** illustrates a schematic diagram of the battery pack replacement system for vehicles in the second embodiment of the present invention.

[0029] FIG. **3** illustrates a schematic diagram of the battery pack replacement system for vehicles in the third embodiment of the present invention.

[0030] FIG. **4** illustrates a schematic diagram of the battery pack replacement system for vehicles in the fourth embodiment of the present invention.

[0031] FIG. **5** illustrates a schematic diagram of the battery pack replacement system for vehicles in the fifth embodiment of the present invention.

[0032] FIG. **6** illustrates a schematic diagram of the battery pack replacement system for vehicles in the sixth embodiment of the present invention.

[0033] FIG. **7** illustrates a schematic diagram of the battery pack replacement system for vehicles in the seventh embodiment of the present invention.

DESCRIPTION OF REFERENCE NUMERALS IN THE DRAWINGS

[0034] **10, 10A, 20, 30, 40, 50, 60**: battery pack replacement system; [0035] **11a, 11b, 11c, 11d, 21a, 21b, 21c, 21d, 31a, 31b, 31c, 31d, 41a, 41b, 41c, 51a**, [0036] **51b, 51c, 61**: first passage; [0037] **12a, 12b, 22a, 22b, 32a, 32b, 42, 52a, 52b, 62**: second passage; [0038] **13a, 13b, 23a, 23b, 33a, 33b, 43a, 43b, 53a, 53b, 63**: processing equipment; [0039] **24, 34a, 34b**: third passage; [0040] **34a1, 34a2, 34b1, 34b2**: third sub-passage; [0041] **15a, 15b, 15c, 15d, 25a, 25b, 25c, 25d, 65a, 65b**: vehicle; [0042] **16, 26, 36, 46, 56a, 56b, 66**: battery pack storage area; [0043] **17a, 17b, 27, 37a, 37b**: battery pack conveyor device; [0044] **37a1, 37a2, 37b1, 37b2**: battery pack sub-conveyor device; [0045] **P11, P12, P13, P14, P15, P16, P17, P18, P21, P22, P23, P24, P25, P26, P27, P28, P41, P42, P43, P44, P45, P46, P61, P62, P63**: processing area.

BEST EMBODIMENT FOR IMPLEMENT THE INVENTION

Best Embodiment of the Invention

[0046] In order to enable those skilled in the art to understand the contents of the present invention and to implement the contents of the present invention accordingly, appropriate embodiments and appended drawings are described as follows, wherein the same components are described with the same component symbols.

[0047] In this embodiment, the defined terms are as follows, where “access” includes place into, take out, and their combinations. The term “finished product” represents an item that is completed in manufacturing and can operate independently, for example, “the EV” as mentioned in this embodiment is a finished product, having at least necessary components such as a battery pack.

[0048] Please refer to FIG. **1**, a battery pack replacement system **10** for vehicles in the first embodiment of the present invention includes four first passages **11a-11d**, two second passages **12a-12b**, two processing equipment **13a-13b**, and a battery pack storage area **16**.

[0049] The four first passages **11a-11d** are spaced apart from each other and located on a first plane, each providing a first object to move within. The first object may be a finished product, such as the EVs **15a-15d**. Taking the EV **15a** as an example, it may be transported along the first passage **11a** using a moving device. The moving device can be an Automated Guided Vehicle (AGV) or an automatic transfer track, for instance. In other embodiments, the vehicle itself can serve as the moving device and move within the first passage through the control of the driver.

[0050] The two second passages **12a-12b** are located on a second plane, different from the first plane. In this embodiment, the second plane is positioned below the first plane. The second passages **12a-12b** intersect with the first passages **11a-11d** in a projection direction, meaning they

partially overlap. Each intersection position (or called overlapping position) of the second passages **12a-12b** with the first passages **11a-11d** has a processing area. In this embodiment, there are a total of eight processing areas **P11-P18**.

[0051] It is worth noting that, in other embodiments, the first passages and the second passages may partially be located on the same plane. The different planes may be formed only in the processing areas. For example, the second passages may be below the first passages only in the processing areas, while in other locations, the second passages and the first passages are in the same plane.

[0052] The two processing equipment **13a-13b** move along their respective second passages **12a-12b** between the corresponding processing areas **P11-P18**. The processing areas **P11-P18** provide positions for the processing equipment **13a-13b** to perform the battery pack replacement operation on the EVs **15a-15d**. For example, the processing equipment **13a** moves along the second passage **12a** to the processing area **P11**, where it performs the battery pack replacement operation on the EV **15a**. The battery pack replacement operation may include the processing equipment **13a** unloading a used battery pack from the EV **15a** and loading a charged battery pack onto the EV **15a**.

[0053] In addition, the second passages **12a-12b** extend to the battery pack storage area **16**, and the processing equipment **13a-13b** can also move between the battery pack storage area **16** and the processing areas **P11-P18**. The battery pack storage area **16** stores a plurality of battery packs, including but not limited to swappable battery packs. Swappable battery packs are designed to be used in conjunction with EVs equipped with main battery packs, primarily providing additional capacity for the EVs. Additionally, each swappable battery pack is equipped with a DC/DC converter that is adaptively designed to correspond to the electrical parameters such as capacity or power of each swappable battery pack. This allows the swappable battery packs to charge, discharge, or collaborate with different main battery packs in various vehicles, achieving broad compatibility.

[0054] Moreover, to further increase the versatility of swappable battery packs, a single large-sized battery pack may be divided into multiple small-sized battery packs. Each small-sized battery pack may have different sizes and/or capacities, and the combination of battery packs with different sizes and/or capacities can cover more possibilities, making them suitable for various vehicle designs.

[0055] In the example of the battery pack replacement system for vehicles, as shown in FIG. 1, the battery pack replacement system **10** may be built at a rest area along a road. The road may include, but is not limited to, long-distance roads or highways. An electric vehicle (EV) can enter the battery pack replacement system **10** during its journey to undergo the battery pack replacement operation. The EV may have a configuration with only a single battery pack or a configuration with both a main battery pack and a swappable battery pack. Before entering the battery pack replacement system **10**, the EV is guided to the entrance of the first passage **11b** from a waiting position, and within the battery pack replacement system **10**, the EV may be transported by the aforementioned mentioned moving devices.

[0056] Subsequently, the moving device moves along the first passage **11b** to the processing area **P12** based on a first guidance information, while the processing equipment **13a** moves along the second passage **12a** to the processing area **P12**. The processing equipment **13a** can move to the processing area **P12** from any position along the second passage **12a**. For example, after completing the battery pack replacement operation in the processing area **P14**, the processing equipment **13a** can then move to the processing area **P12**.

[0057] In the processing area **P12**, the level of the second plane is lower than the level of the first plane, so the processing equipment **13a** is positioned below the EV. The processing equipment **13a** first unloads the used battery pack from the EV and then loads the charged battery pack into the EV to complete the battery pack replacement operation. Subsequently, the moving device moves the EV out of the first passage **11b** based on a second guidance information.

[0058] Since the battery packs used in the EV, whether they are main battery packs or swappable

battery packs, are objects weighing several hundred kilograms and operating at voltage levels of several hundred volts, the processing equipment that meets stability, precision, and safety requirements tends to have a relatively high unit cost. In the battery pack replacement system of the present invention, by allowing the processing equipment to perform the battery pack replacement operation on the EV in at least one of the corresponding processing areas corresponding to different first passages, and by arranging the first passages and the second passage on different planes, a single processing equipment can be scheduled to perform the battery pack replacement operation on different EVs in different first passages. Consequently, the high-cost processing equipment does not need to be fixedly configured for each first passage (i.e., the number of processing equipment can be less than the number of first passages), reducing the unit cost of the battery pack replacement system. When the number of EVs requiring battery pack replacement increases, the processing equipment can be scheduled to move between the corresponding processing areas to increase the utilization rate of the high-cost processing equipment, thereby reducing the unit cost of EV battery pack replacement operations and optimizing the efficiency of the battery pack replacement system. Through such optimization, a balance is achieved between overall investment costs, utilization rates, and operational efficiency. The scheduling method includes, but is not limited to, the processing equipment using the idle time for moving and positioning provided by the EVs on the first passages to immediately perform the battery pack replacement operation on other EVs that have already been positioned on the first passage (or the processing area).

[0059] In addition, FIG. 2 illustrates an application of the battery pack replacement system **10A**, representing the second embodiment of the present invention. In contrast to the battery pack replacement system **10** in the first embodiment, the battery pack replacement system **10A** includes two battery pack conveyor devices **17a-17b**. In the first embodiment, the processing equipment carries both the used battery pack and the charged battery pack, which includes main battery packs and swappable battery packs. However, in the second embodiment, the used battery pack and the charged battery pack are carried by the battery pack conveyor devices **17a-17b**. These battery pack conveyor devices move along the corresponding second passages **12a-12b** between the battery pack storage area **16** and the various processing areas **P11-P18**. The battery pack conveyor devices **17a-17b** access battery packs in the battery pack storage area, and then the processing equipment **13a-13b** collaborates with the battery pack conveyor devices **17a-17b** to perform battery pack access actions. Consequently, the processing equipment **13a-13b** does not need to travel between the processing areas and the battery pack storage area, allowing for a more rapid execution of the battery pack replacement operation.

[0060] It is worth mentioning that, in FIG. 2, both the processing equipment **13a-13b** and the battery pack conveyor devices **17a-17b** move within the second passages **12a-12b**. The configuration of the second passages allows the processing equipment **13a-13b** and the battery pack conveyor devices **17a-17b** to move freely within them without obstructing each other.

[0061] Simultaneously, to further increase the utilization rate, the processing equipment and the battery pack conveyor device can access the required battery packs during their movement. For example, during the movement of the processing equipment towards the next processing area, the unloaded used battery pack can be handed over to the battery pack conveyor device, which is also in motion. The battery pack conveyor device can then acquire the charged battery pack intended for use in the next processing area, effectively utilizing time.

[0062] Next, please refer to FIG. 3. The battery pack replacement system **20**, which is the third embodiment of the present invention, includes four first passages **21a-21d**, two second passages **22a-22b**, two processing equipment **23a-23b**, a third passage **24**, a battery pack storage area **26**, and a battery pack conveyor device **27**.

[0063] The four first passages **21a-21d** are spaced apart from each other in a first plane and are used to transport the EVs **25a-25d**, where each vehicle has at least essential components such as a battery pack (including a main battery pack and/or a swappable battery pack). Taking the EV **25a**

as an example, it can be transported along the first passage **21a** by a moving device. The moving device can be, for example, an Automated Guided Vehicle (AGV) or an automatic transport track. Additionally, the vehicle itself can serve as a mobile device and move within the first passages through driver control.

[0064] The two second passages **22a-22b** are situated in a second plane, different from the first plane. In the third embodiment, the second plane is located below the first plane. In other embodiments, the first passages and the second passages may partially be in the same plane, forming different planes only in the processing area. For instance, the second passage may be below the first passages only in the processing area, while in other locations, the second passage and the first passages are in the same plane.

[0065] The two second passages **22a-22b** intersect with the first passages **21a-21d** in a projection direction. The intersection positions (or called overlapping positions) of the second passages **22a-22b** with the first passages **21a-21d** each have a processing area. In other words, in the third embodiment, there are a total of eight processing areas **P21-P28**.

[0066] The two processing equipment **23a-23b** move along the corresponding second passages **22a-22b** between the corresponding processing areas **P21-P28**. The processing areas **P21-P28** provide locations for the processing equipment **23a-23b** to perform the battery pack replacement operation on the EVs **25a-25d**. For example, the processing equipment **23a** moves along the second passage **22a** to the processing area **P21** and performs the battery pack replacement operation on the EV **25a** in the processing area **P21**.

[0067] The third passage **24** is located on one side of the two second passages **22a-22b**. The term “one side” refers to any angle in the spatial directions of up, down, left, and right, and one end of the third passage **24** extends to the battery pack storage area **26**. The battery pack conveyor device **27** moves along the third passage **24** between the battery pack storage area **26** and the corresponding positions near the processing equipment **23a-23b**. It is worth mentioning that the battery pack conveyor device **27** can obtain the charged battery pack from the battery pack storage area **26** and may acquire the used battery pack unloaded by the processing equipment **23a** or **23b** near the processing area.

[0068] The moving device moves the EV along the first passage **21b** to the processing area **P22**, while the processing equipment **23a** moves along the second passage **22a** to the processing area **P22**. The battery pack conveyor device **27** first obtains a charged swappable battery pack from the battery pack storage area **26** and moves to a position near (or corresponding to) the processing area **P22** to access the battery pack with the processing equipment **23a**. The processing equipment **23a** may unload the used swappable battery pack from the EV first, then load the charged swappable battery pack carried by the battery pack conveyor device **27** onto the EV to complete the battery pack replacement operation. After the processing is complete, the EV is moved out of the first passage **21b** by the moving device.

[0069] The battery pack conveyor device **27** may also include a battery replenishment device and a battery recovery device (not shown in the figure). The battery replenishment device can store charged swappable battery packs, and the battery recovery device can store used swappable battery packs unloaded from the EV. This arrangement provides flexibility in handling charged and used swappable battery packs, contributing to improved operational efficiency.

[0070] In other embodiments, the number of battery pack conveyor devices **27** moving in the third passage **24** is not limited, and the quantity of battery replenishment devices and battery recovery devices is also not restricted. This setup allows for an increased quantity or distance of battery pack transport under the same time conditions, thereby maintaining the processing equipment at maximum efficiency. Additionally, to enhance the efficiency of battery pack transport, multiple battery pack conveyor devices **27** can move within the same third passage **24**. Furthermore, to further improve the utilization rate, the processing equipment and the battery pack conveyor device can access the required battery packs during their movement. For instance, as the processing

equipment moves to the next processing area, it can transfer the unloaded used battery pack to the moving battery pack conveyor device, which can, in turn, acquire the charged battery pack needed for the next processing area, effectively utilizing time.

[0071] Next, please refer to FIG. 4, which illustrates the fourth embodiment of the battery pack replacement system **30** for vehicles in the present invention. It includes four first passages **31a-31d**, two second passages **32a-32b**, two processing equipment **33a-33b**, two third passages **34a-34b**, a battery pack storage area **36**, and two sets of battery pack conveyor devices **37a-37b**. The configurations of the first passages **31a-31d**, second passages **32a-32b**, processing equipment **33a-33b**, and battery pack storage area **36** are the same as those in the third embodiment (the first passages **21a-21d**, the second passages **22a-22b**, the processing equipment **23a-23b**, and the battery pack storage area **26**), and will not be reiterated here.

[0072] The fourth embodiment differs from the third embodiment in that the third passages **34a** and **34b** are adjacent to the second passages **32a** and **32b**. The third passages **34a** and **34b** can be located on the same or different planes as the second passages **32a** and **32b**, for example, on the left and right sides of the same plane, on the upper and lower sides of different planes, or other adjacent orientations. In the fourth embodiment, the third passage **34a** has two third sub-passages **34a1** and **34a2**, each positioned on both sides of the second passage **32a**, and the third passage **34b** has two third sub-passages **34b1** and **34b2**, each positioned on both sides of the second passage **32b**. Additionally, the battery pack conveyor device **37a** has two battery pack sub-conveyor devices **37a1** and **37a2**, which move along the third sub-passages **34a1** and **34a2**, respectively. The battery pack conveyor device **37b** has two battery pack sub-conveyor devices **37b1** and **37b2**, which move along the third sub-passages **34b1** and **34b2**, respectively.

[0073] The battery pack sub-conveyor devices **37a1** and **37b1** may serve as the battery replenishment device described in the third embodiment, while the battery pack sub-conveyor devices **37a2** and **37b2** may serve as the battery recovery device described in the third embodiment. Both the battery replenishment devices and battery recovery devices move between the battery pack storage area **36** and the positions correspond to the processing area. The battery replenishment device provides the charged battery pack to the processing equipment, while the battery recovery device allows the processing equipment to obtain the used battery pack unloaded from the EV. In other embodiments, the battery pack sub-conveyor devices **37a1**, **37a2**, **37b1**, and **37b2** can simultaneously function as both the battery replenishment devices and the battery recovery devices.

[0074] The passages and the battery pack storage area in the above embodiments of the battery pack replacement system can have different configurations, as explained below with reference to FIGS. 5 and 6. It is important to note that these variations can be applied separately or simultaneously to the first through fourth embodiments mentioned above, or in other similar embodiments.

[0075] Please refer to FIG. 5, which illustrates the fifth embodiment of the battery pack replacement system **40** for vehicles in the present invention. It includes three first passages **41a-41c**, one second passage **42**, two processing equipment **43a-43b**, and one battery pack storage area **46**.

[0076] The three first passages **41a-41c** are spaced apart from each other and located on a first plane, used to convey vehicles. The first passages **41a-41c** and vehicles are similar to the embodiments mentioned above, so they will not be described again.

[0077] The second passage **42** is located on a second plane, different from the first plane. In this fifth embodiment, the second plane is below the first plane. The second passage **42** intersects with the first passages **41a-41c** in a projection direction. The intersection positions of the second passage **42** with the first passages **41a-41c** each have a processing area. In other words, in this embodiment, there are a total of six processing areas **P41-P46**. In this embodiment, the second passage **42** is arranged in a U-shape or circular manner in the system, and it can have two or more intersection positions with the first passages **41a-41c**.

[0078] The two processing equipment **43a-43b** move along the second passage **42** between the various processing areas **P41-P46**. In other embodiments, the number of processing equipment can be adjusted according to requirements. In other words, in the same passage, the number of processing equipment can be one or more. Here, the processing equipment **43a-43b** and the battery pack storage area **46** have a similar connection relationship and function to the processing equipment **13a-13b** and the battery pack storage area **16** in the first embodiment, and therefore will not be described again.

[0079] Please refer to FIG. 6, which illustrates the sixth embodiment of a battery pack replacement system **50** for vehicles in the present invention. It includes three first passages **51a-51c**, two second passages **52a-52b**, two processing equipment **53a-53b**, and two battery pack storage areas **56a-56b**. Here, the first passages **51a-51c**, the second passages **52a-52b**, and the processing equipment **53a-53b** have a similar connection relationship and function to the first passages **11a-11d**, the second passages **12a-12b**, and the processing equipment **13a-13b** in the first embodiment. However, unlike the previous embodiments, the two ends of the second passages **52a-52b** extend to the respective battery pack storage areas **56a-56b**.

[0080] In the battery pack storage areas **56a-56b**, it is possible to simultaneously store the charged battery pack, the used battery pack, and their combinations. Alternatively, the battery pack storage area **56a** can store the charged battery pack, while the battery pack storage area **56b** can store the used battery pack, providing different storage configurations.

[0081] Referring to FIG. 7, in the seventh embodiment of the vehicle battery pack replacement system **60**, there is a first passage **61**, a second passage **62**, a processing equipment **63**, and a battery pack storage area **66**. The first passage **61** can accommodate the simultaneous movement of a plurality of EVs **65a-65b**, and the method of movement of the EVs **65a-65b** is as described above, and will not be repeated here.

[0082] The second passage **62** and the first passage **61** have a plurality of overlapping positions in a projection direction, and at each overlapping position, there is a processing areas **P61-P63**. In other words, the second passage **62** and the first passage **61**, in addition to the configuration methods described in the previous embodiments, can also be arranged in a parallel manner. The processing equipment **63** moves along the second passage **62** between these processing areas **P61-P63** to perform the battery pack replacement operation on different EVs. Access to battery packs can be facilitated by the processing equipment **63** accessing the battery pack storage area **66**.

[0083] In the explanation of the seventh embodiment, as shown in FIG. 7, when the EV **65a** moves to the processing area **P62** and completes positioning, the processing equipment **63** performs the battery pack replacement operation on it. Simultaneously, while the EV **65a** undergoes the battery pack replacement operation, the EV **65b** moves to the processing area **P63** and undergoes positioning. After completing the battery pack replacement operation for the EV **65a**, the processing equipment **63** may immediately move to the processing area **P63** and perform the battery pack replacement operation on the EV **65b** that has completed positioning. Therefore, while replacing the battery pack, other EVs waiting for battery pack replacement can simultaneously move and position themselves, saving time that would otherwise be spent waiting. When the number of processing areas is increased, more EVs can move simultaneously, and the efficiency of the system may be further improved by expanding the number of processing equipment.

[0084] In addition, an extended description of the seventh embodiment, the battery pack replacement system may also include a third passage and a battery pack conveyor device (not shown in the figure). The battery pack storage area is adjacent to the third passage, which is located on one side of the second passage. The battery pack conveyor device moves along the third passage between the battery pack storage area and the corresponding positions of the processing equipment. The battery pack conveyor device accesses the battery packs in the battery pack storage area and/or at positions corresponding to the processing equipment. Consequently, the processing equipment can focus on the battery replacement operation.

[0085] Finally, it should be noted that a battery pack replacement system may have a plurality of processing equipment, and some of the processing equipment may be designed for different vehicle specifications or battery pack specifications. Vehicle specifications may include, but are not limited to, dimensions, appearance, wheelbase, track width, ground clearance, and battery pack placement. Battery pack specifications may include, but are not limited to, dimensions, appearance, weight, etc. This design allows the battery pack replacement system to have good scalability and versatility, adapting to various vehicles and battery packs while maintaining efficient scheduling.

[0086] In summary, the battery pack replacement system for vehicles provided by the present invention is designed for replacing battery packs in vehicles, such as electric vehicles (EVs), when their batteries are depleted or for other reasons. The system ensures efficient and balanced operation during peak and off-peak times for battery replacements. The unique configuration involves the first passage and the second passage being arranged in a crossed manner in their projections, allowing high-value (high-priced) processing equipment to move along the second passage. This setup enables the processing equipment to perform battery pack replacement operations on vehicles at the intersection positions. In other words, the first passage and the second passage are not on the same plane, but their projections have crossing points, allowing vehicles and processing equipment to move flexibly on their respective passages. Additionally, since a single processing equipment can perform battery pack replacement operations on EVs on different first passages, the number of second passages (and consequently, the number of processing equipment) can be less than the number of first passages, resulting in lower unit cost while maintaining high utilization and processing efficiency for the entire battery pack replacement system.

[0087] The above description is merely a preferred embodiment of the present invention and should not be construed as limiting the scope of this patent application. Any equivalent modifications or variations made by those skilled in the art, in accordance with the spirit of the present invention, should be included within the scope of the claims.

Claims

1. A battery pack replacement system for vehicles, comprising: a plurality of first passages, which provide a plurality of vehicles to move on them respectively; at least one second passage, which intersects with the first passages in a projection direction, wherein the second passage has a plurality of overlapping positions with the first passages; a plurality of processing areas, which are located in at least part of the overlapping positions; and at least one processing equipment, which is moving along the corresponding second passage between the processing areas, and performing a battery pack replacement operation on at least one of the vehicles in the different processing areas, wherein the battery pack replacement operation includes unloading a used battery pack from the vehicle and/or loading a charged battery pack onto the vehicle by the processing equipment.
2. The battery pack replacement system of claim 1, further comprising: a battery pack storage area, which is adjacent to the second passage and storing a plurality of battery packs, wherein the processing equipment moves between the battery pack storage area and each of the processing areas to transport the used battery pack and/or the charged battery pack.
3. The battery pack replacement system of claim 1, further comprising: a battery pack storage area, which is adjacent to the second passage, and storing a plurality of battery packs; and a battery pack moving device, which is moving along the second passage between the battery pack storage area and the processing equipment, wherein the battery pack moving device accesses the battery packs in the battery pack storage area and/or at a processing equipment corresponding position.
4. The battery pack replacement system of claim 1, further comprising: a third passage, which is located on one side of the at least one second passage; a battery pack storage area, which is adjacent to the third passage, and storing a plurality of battery packs; and a battery pack moving device, which is moving along the third passage between the battery pack storage area and

positions corresponding to the processing equipment, wherein the battery pack moving device accesses the battery packs in the battery pack storage area and/or at a processing equipment corresponding position.

5. The battery pack replacement system of claim 4, wherein the battery pack moving device accesses the battery packs while both itself and the processing equipment are in motion.

6. The battery pack replacement system of claim 1, wherein the number of processing equipment moving along the second passage is more than one.

7. The battery pack replacement system of claim 6, wherein an applicable vehicle specification or an applicable battery pack specification is different for each processing equipment.

8. The battery pack replacement system of claim 1, wherein the battery pack replacement operation further comprises unloading and/or loading a plurality of battery packs for the same vehicle.

9. A battery pack replacement system for vehicles, comprising: a first passage, which provides at least one vehicle to move on it; a second passage, which has a plurality of overlapping positions with the first passage in a projection direction; a plurality of processing areas, which are located in at least part of the overlapping positions; and one or more processing equipment, which being moving along the second passage between the processing areas and performing a battery pack replacement operation on different vehicles within the respective processing areas, wherein the battery pack replacement operation includes the unloading of a used battery pack from the vehicle by the processing equipment and/or the loading of a charged battery pack onto the vehicle by the processing equipment.

10. The battery pack replacement system of claim 9, comprising: a third passage, which is located on one side of the second passage; a battery pack storage area, which is adjacent to the third passage and stores a plurality of battery packs; and a battery pack moving device, which is moving along the third passage between the battery pack storage area and a position corresponding to the processing equipment, wherein the battery pack moving device accesses the battery packs in the battery pack storage area and/or at a processing equipment corresponding position.

11. The battery pack replacement system of claim 10, where in the processing area, the first passage is located on a first plane, the second passage is located on a second plane, and the second plane is below the first plane.

12. The battery pack replacement system of claim 1, where in the processing area, the first passages are located on a first plane, the second passage is located on a second plane, and the second plane is below the first plane.
